Ex 1 (Serie 1 : circuit)

$S = r^2 \pi$ (surface d'un cercle

$d = 2r$

$I = \vec{J} \cdot \vec{n} \cdot S$

$\|\vec{J}\| = j = \frac{I}{S}$

$\vec{J} = n \cdot q \cdot \vec{v}$

$\|\vec{J}\| = n \cdot |q_e| \cdot v_{moy}$

$= j$

Ex 3 $I = \frac{Q}{t}$ $\left(i = \frac{dq}{dt}\right)$

$Q = n |q_e|$

$q_e = -1{,}6 \cdot 10^{-19}\ C$

Ex 6

diviseur de tension →

U_{AB}

A I → R_1 — R_2 — B

u_1 ← u_2 ←

$U_{AB} = u_1 + u_2$

$= R_1 \cdot I + R_2 I$

$U_{AB} = I(R_1 + R_2)$

$u_1 = R_1 \cdot I$

$u_1 = R_1 \cdot \frac{U_{AB}}{R_1 + R_2}$

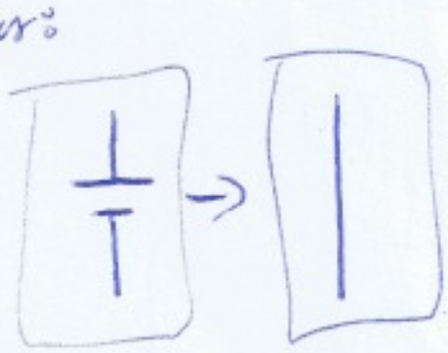
$\Rightarrow \boxed{u_1 = \frac{R_1}{R_1 + R_2} U_{AB}}$ ✗

$\Rightarrow \boxed{u_2 = \frac{R_2}{R_1 + R_2} U_{AB}}$ ✗

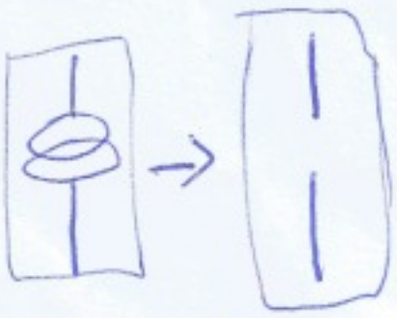
Thevenin :

① - Éteindre les générateurs :

générateur de tension : (court-circuiter)



générateur de courant : (interrupteur ouvert)



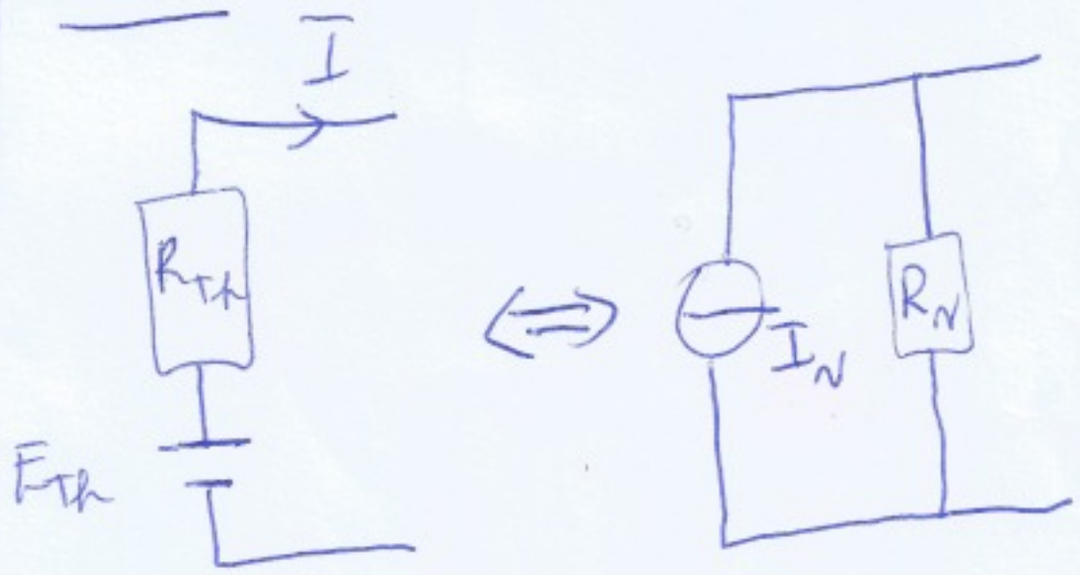
② supprimer (éliminer) la résistance

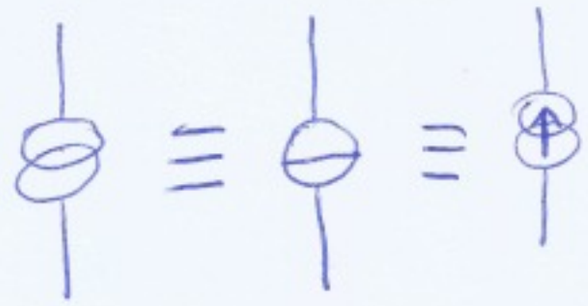
③ calculer R_{Th}

Ex 7

$P = UI$ $U = RI$

Ex 8

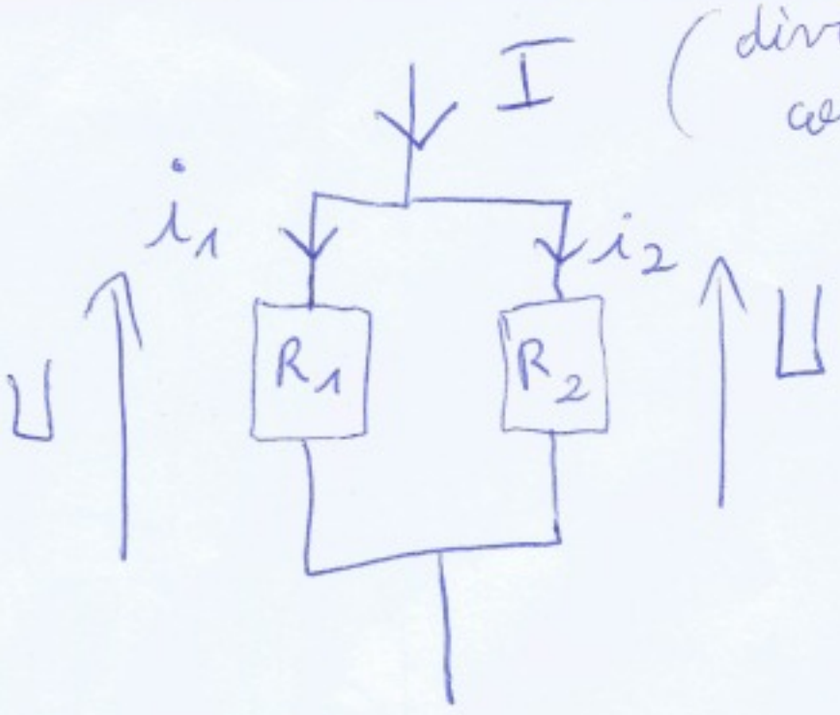




$\begin{cases} R_{Th} = R_N \\ E_{Th} = R_{Th} I_n \end{cases}$ $\Rightarrow I_n = \frac{E_{Th}}{R_{Th}}$

diviseur de tension

(diviseur de courant)



$$I = i_1 + i_2$$

$$\underline{U = R_1 \cdot i_1}$$

$$\underline{U = R_2 \cdot i_2}$$

$$\Rightarrow$$

$$\boxed{i_2 = \frac{R_1}{R_1 + R_2} I} \times$$

$$I = \frac{U}{R_1} + \frac{U}{R_2}$$

$$I = \left(\frac{1}{R_1} + \frac{1}{R_2}\right) U$$

$$i_1 = \frac{U}{R_1}$$

$$i_1 = \left(\frac{I}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)}\right) \frac{1}{R_1}$$

$$\Rightarrow i_1 = \frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2}} I$$

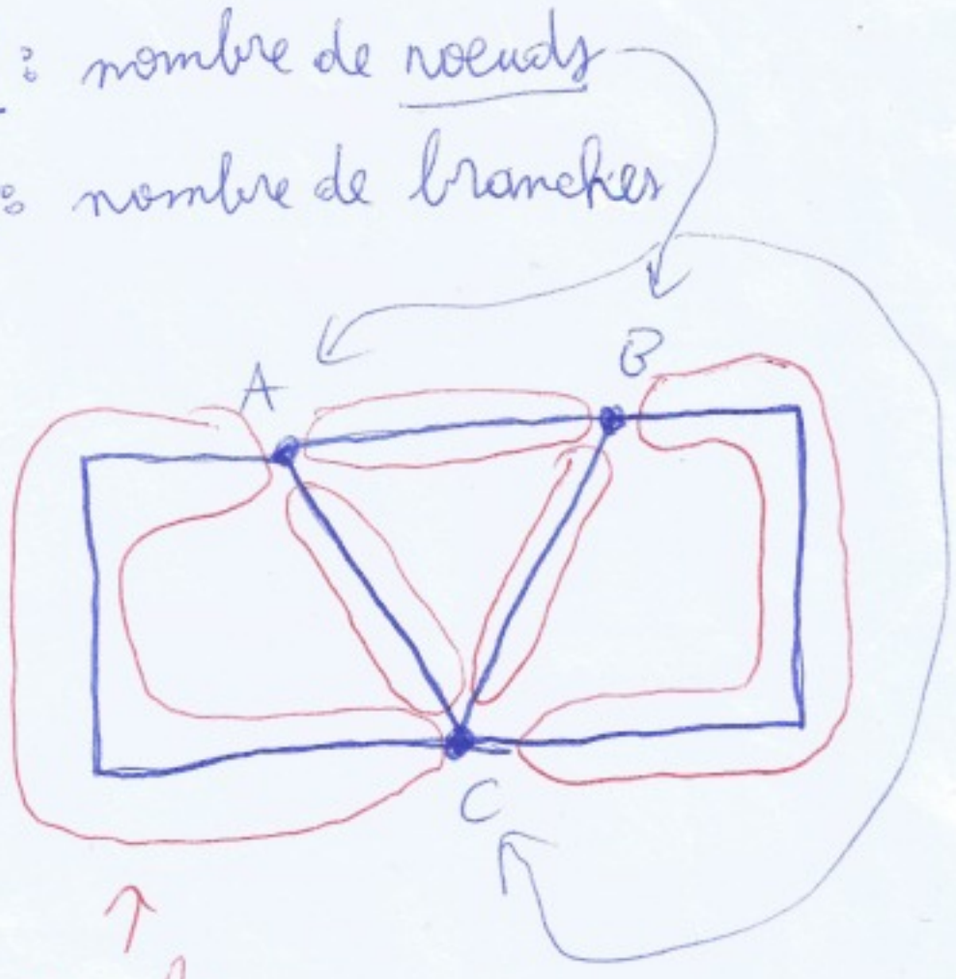
$$= \frac{R_1 R_2}{R_1 R_2} \left(\frac{\frac{1}{R_1}}{(1/R_1) + (1/R_2)}\right) I$$

$$\boxed{i_1 = \frac{R_2}{R_1 + R_2} I} \times$$

Ex 9 :

N : nombre de noeuds

B : nombre de branches



↑ Branche

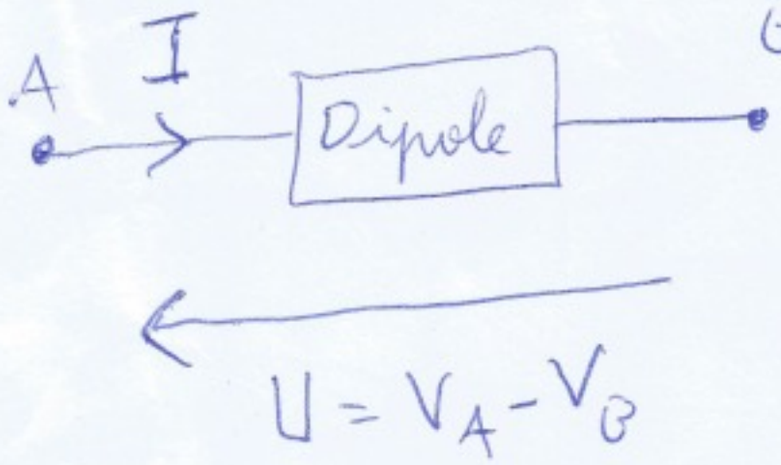
N − 1 : nombre d'équations indépendantes ↑ qui dérivent de la loi des noeuds

$$\boxed{M = B - (N - 1)}$$

M : nombre de mailles et nombre d'équation ↑ qui dérivent de la loi des mailles

B : nombre de branches et nombres d'inconnus

Ex 11: (serie 1 : circuit)



$\Delta R = P \cdot R$

P : pourcentage

(5 % $\Rightarrow$ $P = \frac{5}{100} = 0,05$

$\Rightarrow \Delta R = 0,05 \times R$)

Ex 1: (serie 2 : circuit)

$\begin{cases} U(t) = U_{max} \cdot \sin(\omega t + \varphi) \\ U(t) = U_{max} \cdot \cos(\omega t + \varphi) \end{cases}$

$U_{eff} = \frac{U_{max}}{\sqrt{2}}$

$f = \frac{1}{T}$ (s) (période)

(fréquence) (Hz, s^{-1})

$\omega = 2\pi f$

(pulsation) (rad/s)

$U(t) = U_{max} \cdot \sin(\omega t + \varphi)$

$\overline{U}(t) = U_{max} \cdot e^{j(\omega t + \varphi)}$

$= U_{max} \cdot e^{j\varphi} \cdot e^{j\omega t}$

$\overline{U}_{max} = U_{max} \cdot e^{j\varphi}$

(amplitude complexe)

$\overline{U}(t) = \overline{U}_{max} \cdot e^{j\omega t}$

$\overline{Z} = a + bj$ $\quad \varphi = \arg(\overline{Z}) = \arctg\left(\frac{Im(\overline{Z})}{Re(\overline{Z})}\right)$

$\overline{Z} = |\overline{Z}| \left(\frac{a}{|\overline{Z}|} + \frac{b}{|\overline{Z}|} j\right)$ $\quad \varphi = \arctg\left(\frac{b}{a}\right)$

$\cos(\varphi) = \frac{a}{|\overline{Z}|}$; $\sin(\varphi) = \frac{b}{|\overline{Z}|}$

$\overline{Z} = |\overline{Z}| e^{j\varphi}$

Ex 2: $\overline{Z}_L = jL\omega$ $\quad \overline{Z}_C = \frac{1}{jC\omega}$

$\arctg(+\infty) = +\frac{\pi}{2}$

$\arctg(-\infty) = -\frac{\pi}{2}$

$\arctg(0) = 0$

$\overline{Z}_R = R$

$|\overline{Z}_L| = L\omega \Rightarrow \varphi = \arctg\left(\frac{L\omega}{0}\right) = \frac{\pi}{2}$

$|\overline{Z}_C| = \frac{1}{C\omega} \Rightarrow \varphi = \arctg\left(\frac{-\frac{1}{C\omega}}{0}\right) = -\frac{\pi}{2}$

Ex 3

résonnance : $f = f_0 \Rightarrow I_{eff} = I_{max}$

anti-résonnance : $f = f_0 \Rightarrow I_{eff} = I_{min}$

résonnance $\Rightarrow \omega = \omega_0 \Rightarrow \varphi = 0 \Rightarrow$ I et U en phase

Ex 4

"résistance" maximale $\Rightarrow$ affaiblir le courant

$\Rightarrow$ blocage du passage du courant

$\Rightarrow$ circuit bouchon

Ex 5

$\overline{Z}_L = jL\omega \Rightarrow$ élément inductif

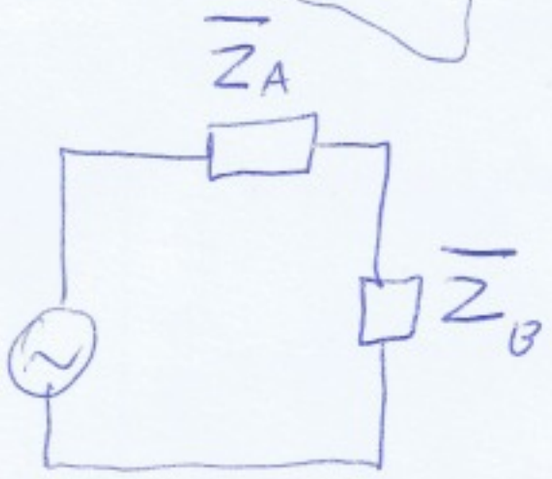
$\overline{Z}_C = \frac{1}{jC\omega} = -j\frac{1}{C\omega}$ élément capacitif

$R < 0$

$R > 0$

$$\bar{Z}_A = \bar{Z}_B^* \leftarrow \text{conjugé} \left(\bar{Z}_B = R_B + jX_B \;;\; \bar{Z}_B^* = R_B - jX_B \right)$$

$\rightsquigarrow$ adaptées

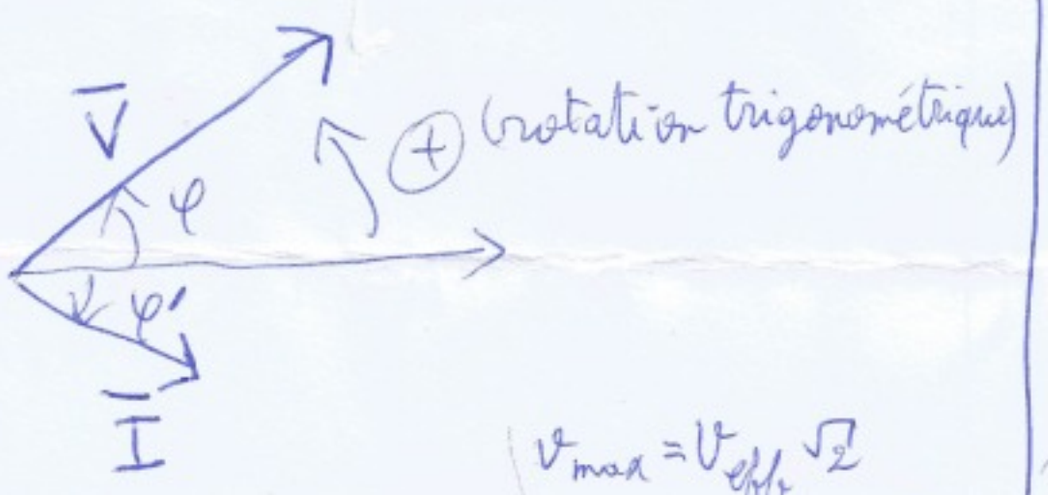


$$\begin{cases} \bar{Z}_A = R_A + jX_A \\ \bar{Z}_B = R_B + jX_B \end{cases}$$

$$\bar{Z}_A = \bar{Z}_B^* \Rightarrow \begin{cases} R_A = R_B \\ X_A = -X_B \end{cases}$$

Ex 6

$$U = RI$$
$$\|\bar{V}\| > \|\bar{I}\|$$



$$v_{max} = v_{eff}\sqrt{2} \qquad I_{max} = I_{eff}\sqrt{2}$$

$$v(t) = V_{max} \sin(\omega t + \varphi)$$
$$i(t) = I_{max} \sin(\omega t + \varphi')$$

$$\Rightarrow \begin{cases} \bar{V} = V_{eff}\sqrt{2}\, e^{j\varphi} \\ \bar{I} = I_{eff}\sqrt{2}\, e^{j\varphi'} \end{cases}$$

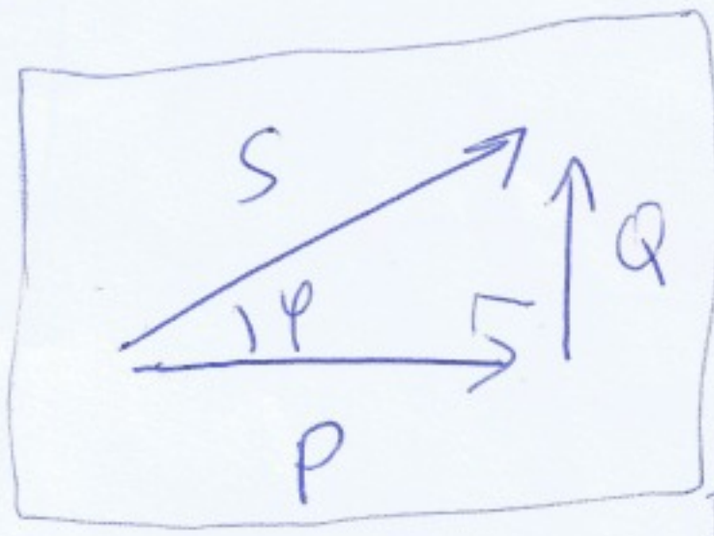
$$\bar{Z} = \frac{\bar{V}}{\bar{I}} = \frac{V_{eff}}{I_{eff}} e^{j(\varphi - \varphi')}$$

$\bar{Z} = \boxed{a} + j\boxed{b}$ ← réactance

(a : résistance)

$\begin{cases} b > 0 \text{ (inductif)} \\ b < 0 \text{ (capacitif)} \end{cases}$

Ex 7 Ex 8



(diagramme des puissances)

$$\tan(\varphi) = \frac{Q}{P}$$

$$\cos(\varphi) = \frac{P}{S} \Rightarrow P = S\cos(\varphi) \times$$

(facteur de puissance)

$$\sin(\varphi) = \frac{Q}{S} \Rightarrow Q = S\sin(\varphi) \times$$

$$S^2 = P^2 + Q^2 \qquad S = \sqrt{P^2 + Q^2}$$

$$S = U_{eff} \cdot I_{eff} \times$$

$$P_{moy} \equiv P_{active} \equiv P_{utile} \equiv \boxed{P} \quad [W] \text{ (watt)}$$

$$P_{réactive} \equiv \boxed{Q} \quad [VAR] \text{ (volt ampère réactive)}$$

$$P_{apparente} \equiv \boxed{S} \quad [VA] \text{ (volt ampère)}$$

$$P = S\cos(\varphi) \qquad Q = S\sin(\varphi)$$
$$P = U_{eff} \cdot I_{eff} \cdot \cos(\varphi) \qquad Q = U_{eff} \cdot I_{eff} \cdot \sin(\varphi)$$

$$I_{eff} = \frac{P}{U_{eff}\cdot\cos(\varphi)} = \frac{Q}{U_{eff}\cdot\sin(\varphi)}$$

Exam normal circuit [Ex1]



$Z = \frac{U}{I}$

φ : déphasage (l'angle entre $i(t)$ et $U(t)$)

[Ex2] $\eta = \frac{P_u}{P_a}$ ← (Puissance Utile)

(rendement) ↑ (Puissance active)

$$\eta = \frac{P_{ut}}{P_{ab}}$$